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| **WSABI Summer Research Lab** | **Summer 2026** |

# Lab 10: Sports Betting

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## 10.1 Introduction

The goal of this lab is to work through the full betting workflow:

price → implied probability → edge → stake size → bankroll consequences.

You will work with a small betting board containing several common market types.

| **Market type** | **Example** |
|---|---|
| Sportsbook moneyline | Sixers `+130` vs Celtics `-150` |
| Point spread | Eagles `-3.5` at `-110` |
| Total | Phillies/Mets over 8.5 at `-105` |
| Prediction market | Contract costs $0.62 and pays $1 if it resolves YES |
| Parlay | Two legs with estimated win probabilities |
| Correlated bets | Same-game side plus player prop |

Throughout the lab, assume a bankroll of $1,000 unless stated otherwise.

## 10.2 Reusable Building Blocks

The starter code provides helper functions for the following tasks:

- Convert American odds to decimal odds.
- Compute the break-even probability $q = 1/d$ from decimal odds $d$.
- For a prediction-market contract with price $c$ and $1 payout, compute its break-even probability and equivalent decimal odds when fees are ignored.
- Compute expected value per $1 staked:
$$EV = dp - 1.$$
- Compute the full Kelly fraction:
$$f^* = \max\left(0, \frac{dp-1}{d-1}\right).$$
- Convert two-sided market prices into no-vig probabilities.

You do **not** need to rewrite those basic helpers unless you want to. Instead, your job is to use them to build the actual analysis tables and simulation code for the parts below.

## 10.3 Part A: Translate Prices into Probabilities

Use the following table.

| Bet | Odds / Price | Model probability |
|---|---|---|
| Team A moneyline | +140 | 0.46 |
| Team B moneyline | -160 | 0.59 |
| Spread bet | -110 | 0.55 |
| Prediction contract | $0.63 | 0.68 |
| Underdog | +250 | 0.31 |
| Favorite | -220 | 0.66 |

### Tasks

1. Convert the sportsbook prices into decimal odds.
2. Compute the break-even probability for every row.
3. Compute the expected value per $1 staked using the supplied model probability.
4. Label each opportunity as positive expected value, negative expected value, or roughly neutral.
5. Rank the bets from best to worst by expected value per $1 staked.

*Betting is not about identifying the more likely winner. It is about identifying when the price is favorable relative to your probability estimate.*

## 10.4 Part B: Remove the Vig

Now consider two-sided markets quoted by a sportsbook.

| Market | Side A | Side B | Your model for Side A |
|---|---|---|---|
| Game 1 | -150 | +130 | 0.60 |
| Game 2 | -110 | -110 | 0.53 |
| Game 3 | +105 | -125 | 0.49 |

### Tasks

1. Convert each side to decimal odds.
2. Compute the raw implied probabilities $q_A$ and $q_B$.
3. Compute the sportsbook hold:
$$h = q_A + q_B - 1.$$
4. Remove the vig by normalizing:
$$\widetilde{q}_A = \frac{q_A}{q_A + q_B}, \qquad \widetilde{q}_B = \frac{q_B}{q_A + q_B}.$$
5. Compare your model probability for Side A to the no-vig probability for Side A. In which markets does your model disagree with the no-vig market the most?

## 10.5 Part C: Kelly Sizing

For every positive-expected-value bet from Part A, compute the full Kelly fraction

$$f^* = \max\left(0, \frac{dp - 1}{d - 1}\right).$$

Then compute the corresponding stake sizes for:

- Full Kelly
- Half Kelly
- Quarter Kelly
- Flat 1% betting

Use a starting bankroll of $1,000.

### Tasks

1. Compute the dollar stake under each of the four staking rules.
2. Rank the positive-expected-value bets by full Kelly fraction.
3. Which bets receive the largest Kelly stakes? Is it always the bet with the highest win probability?

### 10.5.1 Parlays and Correlation

Now add two complications.

1. **Parlay.** Consider a two-leg parlay. Leg 1 has estimated win probability 0.58. Leg 2 has estimated win probability 0.54. The sportsbook offers the parlay at `+290`. Assume the legs are independent.
   (a) Compute the estimated parlay win probability.
   (b) Convert the offered odds to decimal odds and compute the break-even probability.
   (c) Compute the expected value per $1 staked.
   (d) Compute the full Kelly fraction if the parlay is positive expected value.
2. **Correlated bets.** Suppose your model likes Eagles `-3.5` at `-110` and also likes a quarterback passing-touchdown prop at `+120`. Your model probabilities are 0.55 and 0.49, but both bets depend on the same offensive game script.
   (a) Explain why computing separate Kelly stakes for each bet can overstate the appropriate total exposure.
   (b) Give one conservative way to adjust for correlation in practice.

## 10.6 Part D: Simulate Bankroll Growth

In this part, study how staking rules affect long-run bankroll growth.

Build a small board of *independent* opportunities using the positive-expected-value bets from Part A and the parlay from Part C if it is positive expected value. For the simulation, treat the supplied model probabilities as the true probabilities.

Simulate 1,000 bankroll paths of length 500 bets.

At each step, sample one bet at random from your positive-expected-value betting board and settle it according to its win probability.

Compare the following strategies:

- Flat stake: bet 1% of bankroll on every positive-expected-value bet.
- Full Kelly.
- Half Kelly.
- Quarter Kelly.
- Reckless flat stake: bet 10% of bankroll on every perceived positive-expected-value bet.

### Tasks

1. Plot the median bankroll over time for each strategy.
2. Plot the 10th and 90th percentile bankroll paths.
3. Estimate the probability that each strategy loses at least 50% of bankroll at some point.
4. Compute the geometric mean final bankroll for each strategy.
5. Which strategy grows fastest? Which strategy has the worst drawdowns? Which strategy would you actually use?

## 10.7 Part E: Model Uncertainty and Overbetting

Kelly is only optimal relative to the probabilities you feed into it. To study this, introduce model error.

Let $p_i$ denote the true win probability for bet $i$, and let the model probability be

$$\widehat{p}_i = p_i + \varepsilon_i,$$

where $\varepsilon_i$ is mean-zero noise. In code, you may draw

$$\varepsilon_i \sim N(0, 0.03^2)$$

and then truncate $\widehat{p}_i$ to stay inside $[0.01, 0.99]$.

Use the true probabilities to simulate outcomes, but size bets using $\widehat{p}_i$ instead of $p_i$.

### Tasks

1. Repeat the bankroll simulation from Part D using noisy probability estimates.
2. Compare full Kelly, half Kelly, and quarter Kelly under model error.
3. Which staking rule is most sensitive to overconfident probability estimates?
4. When the model is noisy, which strategy appears most robust?

## References

[JLK] Kelly, J. L., *A New Interpretation of Information Rate*, Bell System Technical Journal, 1956.